

IELTS Preparation & Analytics Mobile App

Comprehensive System Architecture, Module Lifecycle & Presentation Documentation

Framework: Flutter (Dart 3.x) & GetX **Storage:** Local-First (SharedPreferences) **Audience:** Project Supervisor / Faculty

1. Executive Summary (প্রকল্পের সারসংক্ষেপ)

প্রেজেন্টেশন ওপেনিং স্টেটমেন্ট: "Sir, this application is a comprehensive, local-first IELTS preparation and real-time candidate analytics mobile solution designed to provide end-to-end practice across all 4 IELTS skill modules (Listening, Reading, Writing, Speaking) with Cambridge-standard automated band score estimation and dynamic performance tracking."

2. System Architecture & Technology Stack

Frontend & UI Layer

Built with **Flutter SDK** and **flutter_screenutil** ensuring pixel-perfect responsive layouts across any screen density with custom Material 3 teal-slate design language.

State Management & DI

Powered by **GetX** for reactive state updates (`.obs`, `Obx`) and clean decoupled Dependency Injection (`Get.put`, `Get.find`).

Persistence & Offline Engine

Engineered with a local-first caching architecture via **SharedPreferences** preserving user test history, band trends, and candidate goals without network dependencies.

Audio Processing & Streams

Integrated with **just_audio**, **audio_waveforms**, and **waveform_flutter** enabling live audio streaming, microphone recording, and interactive visual waveforms.

3. Application Flow & User Journey (কাজের ধারা)

[Application Launch] | ▼ [Splash Controller] → (Reads persistent local state flags) |
|— Case 1: First-Time User —→ [Onboarding Screen] → [Candidate Setup Screen]
|— Case 2: Incomplete Setup —→ [Candidate Setup Screen] (Sets Target Band & Exam Date)
|— Case 3: Registered Candidate —→ [Main Bottom Navigation Hub (5 Core Tabs)]

4. Core Modules Breakdown (মূল ৫টি মডিউল)

Tab / Module	Core Responsibilities	Key Technical Highlights
1. Dashboard HomeScreen	Candidate overview, daily 4-skill checklist, vocabulary booster, and band score calculator launcher.	Reactive daily checklist state syncing with local storage; instant target band progress banner.
2. 4 Skills Practice ScriptScreen	Modular practice engine for Listening, Reading, Writing, and Speaking tasks.	Includes timed passages, live word counter for essays, audio streaming player, and cue cards.
3. Mock & Audition AuditionScreen	Speaking exam simulator with audio recording and audition review.	Live microphone waveform rendering, real-time timer, and local audio persistence.
4. Community CommunityScreen	Collaborative study resource sharing and educational drive access.	Integrated external resource launchers and categorized topic feeds.
5. Analytics SettingScreen	Skill-by-Skill Performance dashboard, Cambridge score calculators, and goal reconfigurator.	Live-updating accuracy progress bars and official Cambridge band score rounding engine.

5. Analytics & Scoring Engine Deep Dive (ডাইনামিক স্কোরিং লজিক)

The app's performance tracking (**Skill-by-Skill Performance**) is completely dynamic and recalculates across all modules in real-time:

 Listening Tracks score accuracy against total questions and estimates Band (0 - 9.0).	 Reading Calculates passage score percentages and logs historical attempts.	 Writing Monitors submission counts, word length compliance, and rubric bands.	 Speaking Logs cue card practice consistency and estimated fluency metrics.
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📐 Official Cambridge Band Score Rounding Formula:

$$\text{Average} = (\text{Listening} + \text{Reading} + \text{Writing} + \text{Speaking}) / 4.0$$

- If fractional remainder < 0.25 → Rounded down to nearest whole band.
- If fractional remainder ≥ 0.25 and < 0.75 → Rounded to half band (.5).
- If fractional remainder ≥ 0.75 → Rounded up to next whole band.

6. Key Highlights & Strengths for Presentation (স্যারকে বলার জন্য বিশেষ পয়েন্ট)

1. Local-First & Zero Latency Users can practice listening audios, reading passages, and study offline without interruptions or costly server latencies.	2. Reactive Data Architecture Every completed quiz immediately broadcasts changes through GetX reactive streams without needing manual page reloads.
3. Official IELTS Rubric Alignment Includes official Cambridge band descriptors and calculation logic, making it pedagogically accurate for actual test takers.	4. Future Extensibility (AI Ready) Modular controller structure allows seamless integration with AI APIs (like Gemini for automatic essay grading and speech analysis).

7. Conclusion & Next Milestones

This project demonstrates a production-ready Flutter mobile architecture blending state-of-the-art state management, audio hardware integration, and pedagogical testing algorithms to create a seamless IELTS test-preparation journey.